

# AXIA NIST AI RMF 1.0 + Generative AI Profile Crosswalk — v1.0

**Document owner:** Yohay Etsion (Fractional CPO, compliance sponsor) **Technical controls attested by:** Dan (AI lifecycle owner), Orr (Head of Engineering) **Regulatory framing reviewed by:** @general-counsel (pending Pass 2 sign-off) **Independence verification:** @ai-architect (controls mapping accuracy), @chief-architect (independence), @general-counsel (regulatory framing) **Version:** 1.0 **Date:** 2026-04-20 **Status:** Draft for verification (Phase 2 deliverable per Valley Bank Master Response Plan)

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**Jurisdiction Assumed:** U.S. federal + NJ (Valley National Bank) for Wrapper framing; CORE is jurisdiction-neutral. If your jurisdiction differs, treat every finding below as a hypothesis to verify with local counsel and the bank's vendor-risk function.

## Architecture Context

"AXIA ships two components. The Gatekeeper and full system run locally under customer control with no egress. LLM-based severity judgment consumes an inference endpoint provisioned per engagement via one of three modes — cloud-infra-hosted (customer's contracted service), customer's internal LLM, or AXIA-provided (Llama 3.3 70B default, alternate open-weights swap path). The AI risk surface is resident with whichever party operates the endpoint."

*Anchor: Architecture Lock Memo v1.0 §D2 (verbatim boilerplate). The three-mode framing (Mode (a)/(b)/(c)) is authoritative for every row in this Crosswalk that references "the LLM," "the severity-judgment LLM," or "AXIA's LLM" — each such reference must be read as "the LLM endpoint consumed for Tier-2 severity judgment (provisioned per engagement via Mode a/b/c)." The AI risk surface allocation (Lock Memo §D1) is the lens through which the Maturity tags in §§2–3 are read.*

## Valley Wrapper (Context)

This document is submitted to Valley National Bank in support of vendor AI questionnaire item **K.6 (Validation and monitoring for regulatory compliance)** as supporting material. The CORE below is bank-agnostic and applies to every AXIA customer deployment; Valley-specific framing is confined to this Wrapper and does not modify the CORE. AXIA commits to honest current-state disclosure: where a control is shipped, it is tagged **Implemented** with an evidence pointer; where it is partial, the specific gap is named; where it is planned, the target is stated without creating a binding delivery warranty. Nothing in this Crosswalk constitutes a claim of NIST AI RMF conformance; the sole purpose is to map AXIA's existing and planned controls to named NIST Functions and Actions so Valley's MRM and vendor-risk teams can place AXIA in their own compliance matrix.

## 1. Framing Statement (CORE)

AXIA is **designed with reference to** NIST AI RMF 1.0 (Govern / Map / Measure / Manage) and the NIST Generative AI Profile (NIST AI 600-1, July 2024). This crosswalk is a mapping of AXIA's existing controls to NIST Functions and Actions, **not an**

attestation.

**AXIA does not claim NIST AI RMF conformance.** No customer, regulator, or assessor should rely on this table as a conformance claim. The crosswalk is provided to support Valley Bank's own vendor-risk and model-risk-management functions in placing AXIA within the bank's compliance matrix. It is not a substitute for the bank's SR 11-7 effective challenge, for independent third-party assessment, or for any attested control framework (e.g., SOC 2, ISO/IEC 42001).

AXIA is also **designed to support customer SR 11-7 obligations**. SR 11-7 (Federal Reserve Supervisory Letter 11-7, 2011; OCC Bulletin 2011-12) applies to supervised financial institutions, not to AXIA. AXIA does not claim SR 11-7 compliance; AXIA produces the evidence substrate (model documentation, explainability artifacts, monitoring telemetry, change logs) that a customer MRM team needs to execute effective challenge against AXIA as a vendor model.

1.1 Scope

The Crosswalk covers AXIA's insider-risk classification pipeline as documented in PRD V2 (§16.8, §16.9, §16.10, §18.3, §21.4), the AXIA AI system architecture ( `axia-ai-system-analysis.md` ), the V4 severity-classification pipeline ( `v4-pipeline-analysis.md` ), the Severity Classification Framework ( `severity-classification-framework.md` ), and the Phase 1 policy artifacts (SDLC Policy v1.0, Model Monitoring & Lifecycle SOP v1.0, Bias Program Framing v1.0). It maps AXIA controls to NIST Functions and named Actions across all four Core Functions and calls out GenAI-Profile-specific actions where applicable.

1.2 Three-State Maturity Tagging

Every mapped control is tagged with one of three maturity states:

| Tag               | Meaning                                                                                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implemented       | Shipped today. Evidence pointer cites a specific PRD section, architecture document, or Phase 1 policy artifact. The control is live and observable.                                                            |
| Partial           | Partially in place. The structural commitment exists in product architecture or contract, but operational artifact (SOP, cadence documentation, first-run output) is still maturing. The specific gap is named. |
| Planned-with-date | Roadmap item with a target period. No binding delivery warranty is created by this Crosswalk; target is a planning signal subject to AXIA's Head of Product and CEO.                                            |

1.3 Honest Positioning on Govern-Function Maturity

Govern is the AXIA function with the most gaps. As a venture-stage AI vendor (founded 2024, current headcount ~8, Seed stage), AXIA does not yet have a formal written AI Policy, a standing AI Ethics Board, a published AI risk-tolerance statement, or a board-level AI oversight charter. The Crosswalk maps these correctly to **Planned** rather than shading them into **Implemented**. This is the correct venture-stage posture; banks see AI vendors constantly, and overclaim is the most common reason a vendor is disqualified during diligence.

1.4 NIST Action-ID Verification Status

AXIA does not warrant that the action-IDs cited in this Crosswalk are complete or exact against the published NIST AI RMF 1.0 Core and the NIST Generative AI Profile text. Action-IDs have been mapped at the family level based on the NIST AI RMF Core structure as understood in the public record. Where an identifier is cited as "pending published-text cross-reference," the Reviewer Checklist requires verification against the NIST publications before this Crosswalk is relied on as a binding submission. The Bias Program Framing v1.0 (Section 8) cited preliminary action-IDs as "mapping hypotheses pending verification during Phase 2.1"; that verification step is this document, and the verification status of each cited identifier is captured in Section 5 below.

1.5 Forward-Looking Statements — No Warranty

Entries in this Crosswalk tagged "Planned" or "Planned-with-date," together with any target dates, roadmap milestones, or forward-looking statements in this document, are planning signals, not contractual delivery commitments. Nothing in this Crosswalk modifies representations, warranties, or commitments made in the executed Master Services Agreement or

Enterprise License Agreement v2, and in case of conflict between this Crosswalk and a customer agreement, the customer agreement controls. Where this Crosswalk is incorporated by reference into a customer agreement, the incorporation applies to structural mappings and Implemented-status rows, and does not extend to Planned or target-dated entries.

### 1.6 Input Minimization as Architectural Governance Commitment (GV-1.3 primary; MP-2.3 secondary)

**Input minimization as an architectural governance commitment (GV-1.3).** AXIA's AI system does not ingest protected-class attributes, demographic data, or HR personnel records (codified in PRD V2 §18.3: Data Minimization, No Employee Scoring, Metadata-Only Mode, Cross-Tenant Isolation). This is a strategic governance choice — codified in Bias Program Framing Layer 1 and reinforced by DPA §9.5.4 (no automated decisions with legal or similarly significant effects on individuals) — that scopes what AXIA refuses to ingest regardless of customer configuration. Secondary mapping: MP-2.3 (data minimization in the Map function). Maturity: **Implemented** (architectural).

This dedicated paragraph upgrades what prior drafts carried as a one-sentence callout. The governance-function mapping is made explicit here because input minimization is AXIA's strongest structural bias control and sits at the Govern-function level (a policy-and-posture commitment scoping the system's input surface), with a secondary Map-function pointer (MP-2.3 data minimization in the Map function). The GV-1.3 and MP-2.3 matrix rows below carry a cross-reference back to this section.

### 1.7 GLBA / Financial-Services Data Handling Posture (GV-1.1 anchor)

Where this Crosswalk's Govern function touches GLBA or financial-services data-handling obligations (primarily GV-1.1 "legal / regulatory requirements regarding AI are understood, managed, and documented"), the authoritative framing is the Architecture Lock Memo §D3 corrected §314.4(f) paragraph, reproduced verbatim below:

"AXIA complies with the FTC Safeguards Rule at 16 CFR §314.4 as a service provider processing NPI, including the administrative, technical, and physical safeguards specified therein. Our contractual framework, compensating controls, and access-management posture address §314.4(f) service-provider oversight requirements. We do not make consumer credit decisions and are not a financial institution for GLBA privacy-notice purposes under 16 CFR Part 313."

*Anchor: Architecture Lock Memo v1.0 §D3 (verbatim). This paragraph replaces any earlier §314.4(f) framing, including any residual "GLBA applies to Valley, not AXIA" phrasing from prior drafting cycles (retired per the VNB Specificity Ladder rubric Finding 3). All Govern-function rows below that touch GLBA, NPI, or consumer-financial-data obligations read against this paragraph; Measure and Manage rows that touch NPI-handling telemetry (e.g., MS-2.10 privacy risk, MG-3.1 third-party risk) also read against this paragraph.*

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## 2. NIST AI RMF 1.0 Core — Crosswalk Matrix (CORE)

The matrix is organized by NIST Function (Govern / Map / Measure / Manage), with a separate Generative AI Profile section in §3. Within each Function, rows are organized by action family. The "Action ID" column cites the identifier at the family level; exact sub-action identifiers are listed where verifiable, otherwise noted as family-level references pending verification.

## 2.1 GOVERN — Organizational Policies, Processes, and Accountability

| Action ID | Action Family                                                                                                                                | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                               | Evidence Pointer                                                                                                                                                       | Maturity                                                                                                                                                                                           | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GV-1.1    | Legal / regulatory requirements regarding AI are understood, managed, and documented                                                         | Data minimization policy (no protected-class ingest); purpose limitation for insider-risk use; DPA §9.5.4 (no automated decisions); DPA §9.5.1 (no training on customer data); FTC Safeguards Rule §314.4 service-provider posture for NPI (per §1.7 verbatim Lock Memo D3 paragraph)                                                 | PRD V2 §18.3 (Privacy by Design); EULA v2 DPA §9.5.1, §9.5.4; Bias Program Framing §2 (Layer 1); <b>Architecture Lock Memo v1.0 §D3 via §1.7</b>                       | <b>Implemented</b>                                                                                                                                                                                 | Data minimization, no-demographic-ingest, and §314.4(f) service-provider posture are load-bearing structural controls. AXIA is not a financial institution for 16 CFR Part 313 privacy-notice purposes. Formal written AI Policy as a standalone artifact is <b>Planned</b> (see GV-1.2 below).                                                                                                                                                                     |
| GV-1.2    | Characteristics of trustworthy AI are integrated into organizational policies, processes, and procedures                                     | No standalone written AI Policy document; trustworthy-AI principles are embedded in PRD V2 §16.10 Safety Guardrails and §18.3 Privacy by Design                                                                                                                                                                                       | PRD V2 §16.10, §18.3                                                                                                                                                   | <b>Planned (target Q4 2026 — post-SOC 2 scoping)</b>                                                                                                                                               | Venture-stage AXIA does not yet have a formal written AI Policy. The principles are present in product documentation; the consolidation into a standalone policy is a planned artifact.                                                                                                                                                                                                                                                                             |
| GV-1.3    | Processes and procedures are in place to determine the needed level of risk management activities based on the organization's risk tolerance | No published AI risk-tolerance statement at the organizational level. Case-level risk framing exists in Severity Classification Framework. <b>Input minimization as architectural governance commitment — see §1.6 (primary mapping)</b> . AI risk surface allocation is per-engagement and is determined by which party operates the | <a href="#">severity-classification-framework.md</a> ; Bias Program Framing §9.5 (change management); <b>§1.6 dedicated paragraph; Architecture Lock Memo v1.0 §D1</b> | <b>Planned (target Q2 2027) for risk-tolerance statement / Implemented (architectural) for input-minimization commitment per §1.6 and per-engagement risk-surface allocation per Lock Memo §D1</b> | Org-level risk-tolerance statement requires board-level articulation and is a post-Series-A artifact. Case-level risk framing is operational today. Input-minimization governance commitment (primary GV-1.3 mapping per §1.6) is Implemented at the architectural level. Under Modes (a) and (b) the AI risk surface for the Tier-2 inference tier is resident with the customer or customer's contracted cloud provider; under Mode (c) it is resident with AXIA. |

| Action ID     | Action Family                                                                                                                                                                                | AXIA Control / Artifact                                                                                                                                                         | Evidence Pointer                                                                                     | Maturity                                          | Notes                                                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                                                                                                                                                                              | inference endpoint (Mode a/b/c per Architecture Lock Memo §D1).                                                                                                                 |                                                                                                      |                                                   |                                                                                                                                                         |
| <b>GV-1.4</b> | The risk management process and its outcomes are established through transparent policies, procedures, and other controls                                                                    | Model Monitoring & Lifecycle SOP v1.0 documents monitoring process, thresholds, retraining triggers, and incident runbook. SDLC Policy v1.0 documents change classes and gates. | Model-Monitoring-Lifecycle-SOP-v1.0.md §§1–8; SDLC-Policy-v1.0.md §§4–6                              | <b>Implemented (Phase 1 artifact)</b>             | SOP is approved v1.0; production operationalization of continuous-cadence metrics gated on monitoring-dashboard delivery (Phase 1 Finding 6 — P0).      |
| <b>GV-1.5</b> | Ongoing monitoring and periodic review of the risk management process and its outcomes are planned                                                                                           | Monitoring SOP cadence: continuous / weekly / monthly / quarterly; annual red-team; annual gold-standard refresh                                                                | Model-Monitoring-Lifecycle-SOP-v1.0.md §2 (cadence table), §7 (red team)                             | <b>Implemented (policy) / Partial (execution)</b> | Cadence is documented; first-quarter red-team and first analyst-override instrumentation are Phase 1 Findings 3, 5 marked P0/P1 for operationalization. |
| <b>GV-2.1</b> | Roles and responsibilities and lines of communication related to mapping, measuring, and managing AI risks are documented and are clear to individuals and teams throughout the organization | Ownership documented across SDLC Policy (§3), Monitoring SOP (§5), Bias Program Framing (§9.2)                                                                                  | SDLC-Policy-v1.0.md §3; Model-Monitoring-Lifecycle-SOP-v1.0.md §5; Bias-Program-Framing-v1.0.md §9.2 | <b>Implemented</b>                                | Named owners: Dan (AI lifecycle), Orr (engineering/monitoring infra), Yohay (compliance/bias sponsor), Asaf (CEO/escalation).                           |
| <b>GV-2.2</b> | The organization's personnel and partners receive AI risk management training                                                                                                                | No formal AI-risk-management training program for personnel; informal onboarding; no partner training program                                                                   | —                                                                                                    | <b>Planned (target Q3 2026)</b>                   | Venture-stage. Formal training curriculum is a post-SOC-2-scoping artifact.                                                                             |
| <b>GV-2.3</b> | Executive leadership of the                                                                                                                                                                  | Asaf Massuri (CEO) is the                                                                                                                                                       | Model-Monitoring-                                                                                    | <b>Implemented</b>                                | CEO accountability is concretely documented in                                                                                                          |

| Action ID     | Action Family                                                                                                                                                            | AXIA Control / Artifact                                                                                                                                                                  | Evidence Pointer                                                                               | Maturity                        | Notes                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|               | organization takes responsibility for decisions about risks associated with AI system development and deployment                                                         | documented escalation owner for model-degradation incidents and bias program findings; sign-off pattern documented for material changes                                                  | Lifecycle-SOP-v1.0.md §5, §6; Bias-Program-Framing-v1.0.md §9.2, §9.5                          |                                 | policy artifacts, not just asserted.                                                                                                               |
| <b>GV-3.1</b> | Decision-making related to mapping, measuring, and managing AI risks throughout the lifecycle is informed by a diverse team                                              | Small venture-stage team (~8); no standing AI ethics or diversity-of-perspective board                                                                                                   | —                                                                                              | <b>Planned (target Q4 2026)</b> | Cannot be authentically claimed at current headcount. Planned post-Series-A as an Advisory Board artifact.                                         |
| <b>GV-3.2</b> | Policies and procedures are in place to define and differentiate roles and responsibilities for human-AI configurations and oversight of AI systems                      | PRD V2 §16.10 Action Boundaries: "AI may only suggest; execution requires human confirmation." Customer analyst retains adjudication over every alert. DPA §9.5.4 no-automated-decisions | PRD V2 §16.10; EULA v2 DPA §9.5.4                                                              | <b>Implemented</b>              | Structural control; reinforced by contractual commitment.                                                                                          |
| <b>GV-4.1</b> | Organizational policies and practices are in place to foster a critical thinking and safety-first mindset in the design, development, deployment, and uses of AI systems | Named accountable owners for Content Filtering (Dan), bias governance (Yohay), escalation (Asaf)                                                                                         | Bias-Program-Framing-v1.0.md §9.2                                                              | <b>Implemented</b>              | Verified identifier per §5 below.                                                                                                                  |
| <b>GV-4.2</b> | Organizational teams document the risks and potential impacts of the AI technology they design, develop,                                                                 | Input-feature inventory with source attribution; PRD V2 §18.3 data minimization; severity-classification                                                                                 | PRD V2 §18.3; severity-classification-framework.md ; Bias-Program-Framing-v1.0.md §3.1, §6.1.1 | <b>Implemented</b>              | Impact documentation includes acknowledged residual risks (behavioral proxies, reconstruction risk) rather than claiming structural impossibility. |

| Action ID     | Action Family                                                                                                                                                                                   | AXIA Control / Artifact                                                                                                                                                                     | Evidence Pointer                                                                       | Maturity                                                 | Notes                                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|               | deploy, evaluate, and use                                                                                                                                                                       | framework documents<br>adjudication rubric; Bias Program Framing documents<br>residual behavioral-proxy risk                                                                                |                                                                                        |                                                          |                                                                                                                                   |
| <b>GV-4.3</b> | Organizational practices are in place to enable AI testing, identification of incidents, and information sharing                                                                                | Incident Response SOP (separate artifact, K.5); model-degradation runbook in Monitoring SOP §6; red-team exercises quarterly                                                                | Model-Monitoring-Lifecycle-SOP-v1.0.md §6, §7                                          | <b>Partial</b>                                           | Model-degradation runbook Implemented; Incident Response SOP is Phase 4 deliverable not yet shipped (see K.5 gap analysis).       |
| <b>GV-5.1</b> | Organizational policies and practices are in place to collect, consider, prioritize, and integrate feedback from those external to the team that developed or deployed the AI system            | Feedback loop: analyst severity adjustments flow back into monitoring (SOP §8); customer reporting via EULA v2 DPA §11 audit rights; customer-facing explainability enables external review | PRD V2 §16.9 Feedback Loop; Model-Monitoring-Lifecycle-SOP-v1.0.md §8; EULA v2 DPA §11 | <b>Implemented (policy) / Partial (instrumentation)</b>  | Analyst-override instrumentation is Phase 1 Finding 3 P0 — SOP defines the loop; dashboard delivery unlocks production telemetry. |
| <b>GV-5.2</b> | Mechanisms are established to enable the team that developed or deployed AI systems to regularly incorporate adjudicated feedback from relevant AI actors into system design and implementation | Model-lifecycle decision log captures remediation decisions post-incident; Monitoring SOP §6 Step 6 mandates post-incident review with monitoring improvements                              | Model-Monitoring-Lifecycle-SOP-v1.0.md §6 Step 6                                       | <b>Implemented (policy) / Partial (first-run)</b>        | SOP formalizes post-incident review; first full-cycle review pending first production incident + 90-day calibration window.       |
| <b>GV-6.1</b> | Policies and procedures are in place to address AI risks                                                                                                                                        | EULA v2 §4 and DPA §6 Sub-Processors regime: 30-day                                                                                                                                         | EULA v2 §4; DPA §6                                                                     | <b>Implemented (contractual) / Partial (operational)</b> | Contractual regime is Implemented. Vendor Risk Register v1 (K.10 Phase 2 deliverable) and AI Supply                               |

| Action ID     | Action Family                                                                                                               | AXIA Control / Artifact                                                                                                   | Evidence Pointer                                        | Maturity       | Notes                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------|
|               | and benefits arising from third-party software and data and other supply chain issues                                       | advance notice, 15-day objection window, flow-down DPA obligations, full liability for sub-processor acts                 |                                                         |                | Chain Posture Document are Partial — structural commitment exists, published artifact pending.  |
| <b>GV-6.2</b> | Contingency processes are in place to handle failures or incidents in third-party data or AI systems deemed to be high-risk | Model-provider switchover plan referenced in Bias Program and DPA; no documented runbook for foundation-model deprecation | EULA v2 §7.2 (platform updates); DPA §6.4.1 (flow-down) | <b>Partial</b> | Contractual foundation exists; dedicated Model-Provider Switchover Runbook is planned artifact. |



## 2.2 MAP — Context, Purpose, Risk Framing

| Action ID | Action Family                                                                                                                                                                                                   | AXIA Control / Artifact                                                                                                                                                                                                   | Evidence Pointer                                                                                               | Maturity                 | Notes                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------|
| MP-1.1    | Intended purpose, potentially beneficial uses, context-specific laws, norms and expectations, and prospective settings in which the AI system will be deployed are understood and documented                    | PRD V2 defines intended purpose (insider-risk classification for financial-institution and enterprise customers); Brand Story and positioning materials define context; Bias Program Framing §1 states purpose limitation | PRD V2 §1 (Purpose), §3 (Scope); AXIA/Marketing/Branding/brand-story-v1.html ; Bias-Program-Framing-v1.0.md §1 | Implemented              | Purpose-and-context documentation is mature; explicit purpose limitation in Bias Framing. |
| MP-1.2    | Inter-disciplinary AI actors, competencies, skills, and capacities for establishing context reflect demographic diversity and broad domain and user experience expertise, and their participation is documented | Venture-stage team; no formal inter-disciplinary body documented                                                                                                                                                          | —                                                                                                              | Planned (target Q4 2026) | Cannot be authentically claimed at current headcount.                                     |
| MP-1.3    | The organization's mission and relevant goals for the AI are understood and documented                                                                                                                          | Company mission and AXIA goals documented in PRD V2 §1–§3 and brand materials                                                                                                                                             | PRD V2 §1–§3                                                                                                   | Implemented              |                                                                                           |
| MP-1.4    | The business value or context of business use has been clearly defined or — in the case of assessing existing AI systems — re-evaluated                                                                         | Business case captured across PRD V2 and pricing strategy document; Severity Classification Framework documents the business-value decision framework for severity thresholds                                             | PRD V2; Product/pricing-strategy-price-book.md ; severity-classification-framework.md                          | Implemented              |                                                                                           |
| MP-1.5    | Organizational risk tolerances are determined                                                                                                                                                                   | No published organizational risk-tolerance                                                                                                                                                                                | —                                                                                                              | Planned (target Q2 2027) | Coupled to GV-1.3.                                                                        |

| Action ID     | Action Family                                                                                                                                                   | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                                                                                                            | Evidence Pointer                                                                                                                     | Maturity           | Notes                                                                                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                                                                                                                                                 | statement (see GV-1.3)                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                      |                    |                                                                                                                                                  |
| <b>MP-1.6</b> | System requirements are elicited from and understood by relevant AI actors. Design decisions take socio-technical implications into account to address AI risks | PRD V2 §16.10 Safety Guardrails and §18.3 Privacy by Design explicitly address socio-technical implications (human confirmation gates, no automated decisions, metadata-only mode)                                                                                                                                                                                                                                 | PRD V2 §16.10, §18.3                                                                                                                 | <b>Implemented</b> |                                                                                                                                                  |
| <b>MP-2.1</b> | The specific task, and methods used to implement the task, that the AI system will support is defined                                                           | Severity classification: ensemble of SetFit + ModernBERT + NN/XGBoost over embeddings, confidence-weighted soft voting, LLM gating, WHO/WHAT/WHO adjudication rubric. The LLM endpoint consumed for Tier-2 severity judgment is provisioned per engagement via Mode (a) cloud-infra-hosted, Mode (b) customer's internal LLM, or Mode (c) AXIA-provided (Llama 3.3 70B default, alternate open-weights swap path). | axia-ai-system-analysis.md ; v4-pipeline-analysis.md ; severity-classification-framework.md ; <b>Architecture Lock Memo v1.0 §D1</b> | <b>Implemented</b> | Ensemble architecture for the gating and supervised-learning tiers is unchanged across modes; only the Tier-2 inference endpoint varies by mode. |
| <b>MP-2.2</b> | Information about the AI system's knowledge limits and how system output may be utilized and overseen by humans is documented                                   | PRD V2 §16.10 Action Boundaries + §16.8 Explainability: human-confirmation gates on all AI-suggested actions; structured reasoning What / Why It Matters / Why It Is Unusual;                                                                                                                                                                                                                                      | PRD V2 §16.8, §16.10                                                                                                                 | <b>Implemented</b> |                                                                                                                                                  |

| Action ID     | Action Family                                                                                                                                                                                                                                                     | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                                                                                      | Evidence Pointer                                                                                                                                                                                    | Maturity           | Notes                                                                                                                                                                                                                                            |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               |                                                                                                                                                                                                                                                                   | confidence indicators<br>High/Medium/Low per alert                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                     |                    |                                                                                                                                                                                                                                                  |
| <b>MP-2.3</b> | Scientific integrity and TEVV considerations are identified and documented, including those related to experimental design, data collection and selection (e.g., availability, representativeness, suitability), system trustworthiness, and construct validation | TEVV substrate: 150-case ground-truth benchmark across 5 frontier models with per-severity accuracy + QWK 0.920 for ModernBERT (V4); Severity Classification Framework provides adjudication rubric for construct validation; Synthetic Data Multi-Model provenance for representativeness. <b>Data minimization in the Map function — see §1.6 (secondary pointer from GV-1.3 primary).</b> | severity-benchmark-analysis.md ; severity-benchmark-detailed-gt.md ; severity-classification-framework.md ; axia-ai-system-analysis.md §Synthetic Data Multi-Model; <b>§1.6 dedicated paragraph</b> | <b>Implemented</b> | Verified identifier per §5 below. Under-presented in current materials; this Crosswalk is the first consolidated surfacing. MP-2.3 also carries the secondary pointer from GV-1.3 primary for input-minimization governance commitment per §1.6. |
| <b>MP-3.1</b> | Potential benefits of intended AI system functionality and performance are examined and documented                                                                                                                                                                | Business case and PRD V2 articulate intended benefits; severity benchmark quantifies performance                                                                                                                                                                                                                                                                                             | PRD V2; severity-benchmark-analysis.md                                                                                                                                                              | <b>Implemented</b> |                                                                                                                                                                                                                                                  |
| <b>MP-3.2</b> | Potential costs, including non-monetary costs, which result from expected or realized AI errors or system functionality and trustworthiness — as connected to organizational principles — are examined and documented                                             | Cost framing: Bias Program §3.1 (facially-neutral behavioral features and disparate-impact surface); Bias Program §6.1.1 (reconstruction risk); Monitoring SOP §4 (retraining triggers on drift); PRD V2 §16.10 (safety guardrails as cost-mitigation)                                                                                                                                       | Bias-Program-Framing-v1.0.md §3.1, §6.1.1; Model-Monitoring-Lifecycle-SOP-v1.0.md §4                                                                                                                | <b>Implemented</b> |                                                                                                                                                                                                                                                  |

| Action ID     | Action Family                                                                                                                                                                                    | AXIA Control / Artifact                                                                                                                                                                                                     | Evidence Pointer                                                                       | Maturity           | Notes                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------|
| <b>MP-3.3</b> | Targeted application scope is specified and documented based on the system's capability, established context, and AI system categorization                                                       | Insider-risk classification for employee communications in corporate and financial-institution contexts; Bias Program §1 purpose limitation; explicit exclusions (no credit decisioning, no automated employment decisions) | PRD V2 §3; Bias-Program-Framing-v1.0.md §1, §7.4                                       | <b>Implemented</b> |                                                                                |
| <b>MP-3.4</b> | Processes for operator and practitioner proficiency with AI system performance and trustworthiness — and relevant technical standards and certifications — are defined, assessed, and documented | Customer analyst training materials are a Phase 1 Bias Framing roadmap artifact; SOC 2 Type II targeted Q4 2026 as independent certification                                                                                | Bias-Program-Framing-v1.0.md §5 (Customer Enablement); Valley Bank Master Plan Phase 5 | <b>Partial</b>     | Training materials drafted, not yet formally versioned; certification planned. |
| <b>MP-3.5</b> | Processes for human oversight are defined, assessed, and documented in accordance with organizational policies from the GOVERN function                                                          | PRD V2 §16.10 human confirmation gates; DPA §9.5.4 no automated decisions; Customer analyst as adjudication owner                                                                                                           | PRD V2 §16.10; EULA v2 DPA §9.5.4                                                      | <b>Implemented</b> |                                                                                |

| Action ID     | Action Family                                                                                                                                                                                                                                                                | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Evidence Pointer                                                                                                         | Maturity                                                                 | Notes                                                                                                                                                                                                                                                |
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| <b>MP-4.1</b> | Approaches for mapping AI technology and legal risks of its components — including the use of third-party data or software — are in place, followed, and documented, as are risks of infringement of a third-party's intellectual property or other rights                   | Sub-processor regime (EULA v2 §4, DPA §6); Tier-2 inference endpoint provenance is per-engagement via Mode (a/b/c) — under Mode (a) the customer's contracted managed LLM service (e.g., AWS Bedrock, Azure OpenAI, GCP Vertex) operates under the customer's service contract and data-handling terms; under Mode (b) the customer hosts internally and AXIA consumes the endpoint; under Mode (c) AXIA provides Llama 3.3 70B default (alternate open-weights swap path documented); Bias Framing §2 cites PRD V2 §18.3 data minimization as IP/privacy minimization | EULA v2 §4; DPA §6; Bias-Program-Framing-v1.0.md §2; axia-ai-system-analysis.md ; <b>Architecture Lock Memo v1.0 §D1</b> | <b>Implemented (contractual) / Partial (AI Supply Chain Posture Doc)</b> | Verified identifier per §5 below. AI Supply Chain Posture Document is K.10 Phase 2 deliverable. Foundation-model legal-risk allocation follows the mode: Modes (a)/(b) customer-side, Mode (c) AXIA-side, per Lock Memo §D1 risk-surface allocation. |
| <b>MP-5.1</b> | Likelihood and magnitude of each identified risk based on expected use, past uses of AI systems in similar contexts, public incident reports, feedback from those external to the team that developed or deployed the AI system, or other data are identified and documented | Bias Program §3.1 documents specific behavioral-proxy surfaces (caregiver patterns, religious observance, ADA, national origin); §6.1.1 documents reconstruction-risk surface; acknowledged as residual risks with Layer 5 audit as defense-in-depth                                                                                                                                                                                                                                                                                                                   | Bias-Program-Framing-v1.0.md §3.1, §6.1.1                                                                                | <b>Implemented (qualitative) / Partial (quantitative)</b>                | Qualitative risk surfacing Implemented; quantitative likelihood / magnitude pending Layer 5 first audit (Q3 2026 target).                                                                                                                            |

| Action ID | Action Family                                                                                                                                                                              | AXIA Control / Artifact                                                                                                         | Evidence Pointer                                         | Maturity    | Notes |
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| MP-5.2    | Practices and personnel for supporting regular engagement with relevant AI actors and integrating feedback about positive, negative, and unanticipated impacts are in place and documented | Feedback loop (PRD V2 §16.9); customer audit rights (DPA §11); semi-annual customer updates on Bias Program (Bias Framing §6.2) | PRD V2 §16.9; DPA §11; Bias-Program-Framing-v1.0.md §6.2 | Implemented |       |

## 2.3 MEASURE — Testing, Evaluation, Verification, Validation

| Action ID | Action Family                                                                                                                                                     | AXIA Control / Artifact                                                                                                                                                                   | Evidence Pointer                                                                                                                    | Maturity                                         | Notes                                                                                           |
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| MS-1.1    | Approaches and metrics for measurement of AI risks enumerated during the MAP function are selected for implementation starting with the most significant AI risks | Monitoring metrics: gate FN rate, per-tier recall (Critical $\geq 90\%$ , High $\geq 85\%$ ), LLM judge consensus, rolling QWK ( $\geq 0.88$ ), analyst override rate, input/output drift | Model-Monitoring-Lifecycle-SOP-v1.0.md §1, §3                                                                                       | Implemented (policy) / Partial (instrumentation) | Metrics defined; continuous-cadence metrics gated on dashboard delivery (Phase 1 Finding 6 P0). |
| MS-1.2    | Appropriateness of AI metrics and effectiveness of existing controls is regularly assessed and updated, including reports of errors and potential impacts         | Monthly drift review + monthly override review; quarterly red team; annual gold-standard refresh; model-lifecycle decision log captures remediation                                       | Model-Monitoring-Lifecycle-SOP-v1.0.md §2, §6, §7                                                                                   | Implemented (policy) / Partial (first-run)       | First-cycle execution pending 90-day calibration window.                                        |
| MS-1.3    | Internal experts who did not serve as front-line developers for the system and/or independent assessors are involved in regular assessments and updates           | Phase 4.5 Adversarial Review pattern (Pattern 5) used for Bias Program Framing (fresh-context @employment-counsel, @general-counsel tiebreaker); red-team exercises quarterly             | Bias-Program-Framing-v1.0.md Reviewer Checklist (Phase 4.5); Model-Monitoring-Lifecycle-SOP-v1.0.md §7                              | Implemented (pattern) / Partial (execution)      | Pattern established; first red-team exercise pending (Phase 1 Finding 5).                       |
| MS-2.1    | Test sets, metrics, and details about the tools used during TEVV are documented                                                                                   | 150-case ground-truth benchmark; severity-benchmark.py harness; V4 pipeline analysis; severity-classification framework                                                                   | severity-benchmark-analysis.md ; severity-benchmark-detailed-gt.md ; v4-pipeline-analysis.md ; severity-classification-framework.md | Implemented                                      |                                                                                                 |
| MS-2.2    | Evaluations involving human subjects meet applicable requirements (including human subject protection) and are                                                    | Benchmark is synthetic + adjudicated-case corpus; no human-subject research in classical sense; demographic representativeness is                                                         | axia-ai-system-analysis.md §Synthetic Data Multi-Model; Bias-Program-Framing-v1.0.md §6                                             | Partial                                          | Structural substrate in place; population-level representativeness test is Layer 5 roadmap.     |

| Action ID     | Action Family                                                                                                                                                                                                                                            | AXIA Control / Artifact                                                                                                                                                                        | Evidence Pointer                                                                 | Maturity                                                      | Notes                                                                                                                                                                             |
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|               | representative of the relevant population                                                                                                                                                                                                                | Layer 5 audit scope (2026)                                                                                                                                                                     |                                                                                  |                                                               |                                                                                                                                                                                   |
| <b>MS-2.3</b> | AI system performance or assurance criteria are measured qualitatively or quantitatively and demonstrated for conditions similar to deployment setting(s). Measures are documented                                                                       | Severity benchmark quantifies performance (per-tier accuracy, QWK 0.920 V4 ModernBERT, Claude Sonnet 4 at 87.3% with framework); Monitoring SOP thresholds (gate FN <2%, Critical recall ≥90%) | severity-benchmark-analysis.md ; Model-Monitoring-Lifecycle-SOP-v1.0.md §1       | <b>Implemented (offline) / Partial (production telemetry)</b> | Offline benchmark Implemented; production telemetry gated on dashboard.                                                                                                           |
| <b>MS-2.4</b> | The functionality and behavior of the AI system and its components — as identified in the MAP function — are monitored when in production                                                                                                                | Monitoring SOP §1 production metrics (continuous + periodic); model-lifecycle decision log; alerting infrastructure                                                                            | Model-Monitoring-Lifecycle-SOP-v1.0.md §1, §2                                    | <b>Partial</b>                                                | Policy defines; production deployment of continuous-cadence metrics gated on dashboard (Phase 1 Finding 6 P0).                                                                    |
| <b>MS-2.5</b> | The AI system to be deployed is demonstrated to be valid and reliable. Limitations of the generalizability beyond the conditions under which the technology was developed are documented                                                                 | Benchmark documents validity/reliability on 150 adjudicated cases; generalizability limits documented (synthetic-case mix, single-domain, single-language)                                     | severity-benchmark-detailed-gt.md ; axia-ai-system-analysis.md                   | <b>Implemented</b>                                            | Under-presented in current materials; Crosswalk surfaces.                                                                                                                         |
| <b>MS-2.6</b> | AI system is evaluated regularly for safety risks — as identified in the MAP function. The AI system to be deployed is demonstrated to be safe, its residual negative risk does not exceed the risk tolerance, and can fail safely, particularly if made | PRD V2 §16.10 Safety Guardrails; DPA §9.5.4 no automated decisions; Monitoring SOP §4 retraining triggers; §6 incident runbook                                                                 | PRD V2 §16.10; EULA v2 DPA §9.5.4; Model-Monitoring-Lifecycle-SOP-v1.0.md §4, §6 | <b>Partial</b>                                                | Safety controls Implemented; organizational risk tolerance (GV-1.3 / MP-1.5) is Planned, so "does not exceed risk tolerance" criterion is unresolved until that statement exists. |



| Action ID      | Action Family                                                                                                                                                                               | AXIA Control / Artifact                                                                                                                                                                                      | Evidence Pointer                                                                              | Maturity                                 | Notes                                                                                                                                                                                                                                                                 |
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|                | to operate beyond its knowledge limits. Safety metrics reflect system reliability and robustness, real-time monitoring, and response times for AI system failures                           |                                                                                                                                                                                                              |                                                                                               |                                          |                                                                                                                                                                                                                                                                       |
| <b>MS-2.7</b>  | AI system security and resilience — as identified in the MAP function — are evaluated and documented                                                                                        | Red-team quarterly cadence; EULA v2 §7.4 security incident notification; prompt-injection testing scoped for red team (NIST GenAI Profile risk)                                                              | Model-Monitoring-Lifecycle-SOP-v1.0.md §7; EULA v2 §7.4                                       | <b>Partial</b>                           | Cadence scheduled; first exercise pending (Phase 1 Finding 5 P1).                                                                                                                                                                                                     |
| <b>MS-2.8</b>  | Risks associated with transparency and accountability — as identified in the MAP function — are examined and documented                                                                     | Structured reasoning per alert; factor attribution; confidence indicators; 100% audit log coverage is a <b>target per PRD V2 §21.4, not currently attested production reality</b> ; no-black-boxes principle | PRD V2 §16.8 Explainability; PRD V2 §21.4 Trust Metrics                                       | <b>Partial (target not yet attested)</b> | Verified identifier per §5 below. 100% audit-log coverage is Trust Metric <b>target</b> per PRD V2 §21.4; whether achieved in production today or remains aspirational is Cannot Assess Without Engineering Confirmation (carried from Bias Program Framing Finding). |
| <b>MS-2.9</b>  | The AI model is explained, validated, and documented, and AI system output is interpreted within its context — as identified in the MAP function — to inform responsible use and governance | Explainability artifacts (What / Why It Matters / Why Unusual); DPA §9.5.3 contractual commitment to explainability; Source Grounding principle                                                              | PRD V2 §16.8; EULA v2 DPA §9.5.3                                                              | <b>Implemented</b>                       | Verified identifier per §5 below.                                                                                                                                                                                                                                     |
| <b>MS-2.10</b> | Privacy risk of the AI system — as identified in the MAP function — is examined and documented                                                                                              | Data minimization (§18.3); metadata-only mode; no training on customer data (§9.5.1); cross-tenant isolation (§9.5.2); data residency (§4); FTC Safeguards Rule §314.4 service-                              | PRD V2 §18.3; EULA v2 DPA §9.5.1, §9.5.2, §4; <b>Architecture Lock Memo v1.0 §D3 via §1.7</b> | <b>Implemented</b>                       | Strongest cluster of controls in the matrix. NPI handling as §314.4(f) service provider; AXIA is not a financial institution for 16 CFR Part 313                                                                                                                      |

| Action ID      | Action Family                                                                                                                                            | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Evidence Pointer                                                                                                  | Maturity                                                         | Notes                                                                                                                                                         |
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|                |                                                                                                                                                          | provider posture for NPI (per §1.7 verbatim Lock Memo D3 paragraph)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                   |                                                                  | privacy-notice purposes.                                                                                                                                      |
| <b>MS-2.11</b> | Fairness and bias — as identified in the MAP function — are evaluated and results are documented                                                         | Bias Program Framing Layers 1–4 (existing): input minimization, engineering diversity (3-model ensemble, 3-vendor synthetic data), customer-configurable guardrails, explainability as customer audit enablement. Layer 5 (2026 roadmap): population-level disparate-impact audit + reconstruction-risk testing. Bias profile of the LLM endpoint consumed for Tier-2 severity judgment is mode-dependent: Mode (a) inherited from the customer's cloud-managed LLM; Mode (b) owned by the customer as endpoint operator; Mode (c) inherited from the published base model (Llama 3.3 70B or alternate open-weights) with no default fine-tuning on customer data. AXIA's bias controls apply to prompt construction, context windowing, Gatekeeper prompt logic, and post-hoc review workflows across all modes. | Bias-Program-Framing-v1.0.md §§2–6;<br><b>Architecture Lock Memo v1.0 §D1; Mode (c) one-pager (@ai-architect)</b> | <b>Partial (Layers 1–4 Implemented; Layer 5 Planned Q3 2026)</b> | Verified identifier per §5 below. Quantitative fairness-metric selection deferred to 2026 employment-counsel review. Bias-surface allocation tracks the mode. |
| <b>MS-2.12</b> | Environmental impact and sustainability of AI model training and management activities — as identified in the MAP function — are assessed and documented | No environmental-impact assessment documented                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                                                 | <b>Planned (target Q4 2026)</b>                                  | Low-priority for venture-stage; becomes relevant with scale.                                                                                                  |
| <b>MS-2.13</b> | Effectiveness of the employed                                                                                                                            | Post-incident review requires monitoring-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Model-Monitoring-                                                                                                 | <b>Implemented (policy) / Partial</b>                            |                                                                                                                                                               |

| Action ID     | Action Family                                                                                                                                                                                                      | AXIA Control / Artifact                                                                                                                                            | Evidence Pointer                                                                       | Maturity                                                      | Notes                                                                                                                                                                                      |
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|               | TEVV metrics and processes in the MEASURE function are evaluated and documented                                                                                                                                    | effectiveness analysis (SOP §6 Step 6): "did the monitoring metrics detect this, or was it a customer report?"                                                     | Lifecycle-SOP-v1.0.md §6 Step 6                                                        | <b>(first-run)</b>                                            |                                                                                                                                                                                            |
| <b>MS-3.1</b> | Approaches, personnel, and documentation are in place to regularly identify and track existing, unanticipated, and emergent AI risks based on factors such as intended and actual performance in deployed contexts | Monthly drift review; quarterly red team; feedback loop; model-lifecycle decision log; semi-annual customer updates                                                | Model-Monitoring-Lifecycle-SOP-v1.0.md §§2, 7, 8; Bias-Program-Framing-v1.0.md §6.2    | <b>Partial</b>                                                | Policy defines; first-cycle execution pending.                                                                                                                                             |
| <b>MS-3.2</b> | Risk tracking approaches are considered for settings where AI risks are difficult to assess using currently available measurement techniques or where metrics are not yet available                                | Bias Program §6.1.1 explicitly acknowledges reconstruction risk as not-yet-measured; Layer 5 scopes testing methodology                                            | Bias-Program-Framing-v1.0.md §6.1.1                                                    | <b>Implemented (acknowledgment) / Planned (measurement)</b>   | Honest framing: acknowledges surface, plans measurement, does not claim "structurally impossible."                                                                                         |
| <b>MS-3.3</b> | Feedback processes for end users and impacted communities to report problems and appeal system outcomes are established and integrated into AI system evaluation metrics                                           | Customer audit rights (DPA §11); customer analyst override feedback loop; escalation path (Schedule C §C.9 to CEO); DPA §7 incident notification                   | EULA v2 DPA §11, §7; Schedule C §C.9; PRD V2 §16.9                                     | <b>Implemented (customer) / Partial (end-user / employee)</b> | Customer-level channels Implemented; direct end-user (monitored employee) appeal channel is customer's operational responsibility per DPA framing; AXIA provides explainability substrate. |
| <b>MS-4.1</b> | Measurement approaches for identifying AI risks are connected to deployment context(s) and informed through consultation with                                                                                      | Severity Classification Framework developed with product + AI + adjudication input; Bias Program Framing reviewed by @general-counsel, @employment-counsel; future | severity-classification-framework.md ; Bias-Program-Framing-v1.0.md Reviewer Checklist | <b>Partial</b>                                                | External domain-expert consultation on bias metrics is 2026 milestone.                                                                                                                     |

| Action ID | Action Family                                                                                                                                                                                                                                       | AXIA Control / Artifact                                                                                                            | Evidence Pointer                                                             | Maturity | Notes                                          |
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|           | domain experts and other end users                                                                                                                                                                                                                  | Customer MRM Integration Guide (Bias Framing Finding 6)                                                                            |                                                                              |          |                                                |
| MS-4.2    | Measurement results regarding AI system trustworthiness in deployment context(s) and across the AI lifecycle are informed by input from domain experts and relevant AI actors to validate whether the system is performing consistently as intended | Monitoring SOP §8 feedback loop; model-lifecycle decision log; post-incident review pattern                                        | Model-Monitoring-Lifecycle-SOP-v1.0.md §6, §8                                | Partial  | Policy defines; first-cycle execution pending. |
| MS-4.3    | Measurable performance improvements or declines based on consultations with relevant AI actors, including affected communities, and field data about context-relevant risks and trustworthiness characteristics are identified and documented       | Semi-annual customer updates on Bias Program (Bias Framing §6.2); post-incident reviews documented in model-lifecycle decision log | Bias-Program-Framing-v1.0.md §6.2; Model-Monitoring-Lifecycle-SOP-v1.0.md §6 | Partial  |                                                |

## 2.4 MANAGE — Risk Treatment, Prioritization, Monitoring

| Action ID     | Action Family                                                                                                                                                                                                 | AXIA Control / Artifact                                                                                                                                                                                          | Evidence Pointer                                                                               | Maturity           | Notes                                                         |
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| <b>MG-1.1</b> | A determination is made as to whether the AI system achieves its intended purposes and stated objectives and whether its development or deployment should proceed                                             | SDLC Policy gates; Monitoring SOP retraining triggers with Asaf escalation; gold-standard refresh cadence                                                                                                        | SDLC-Policy-v1.0.md §5; Model-Monitoring-Lifecycle-SOP-v1.0.md §4                              | <b>Implemented</b> |                                                               |
| <b>MG-1.2</b> | Treatment of documented AI risks is prioritized based on impact, likelihood, and available resources or methods                                                                                               | Severity tiering in SOP §1 (P0 Critical / High recall thresholds most stringent); retraining-trigger priority by impact (14-day investigation for Critical recall, 30-day for input drift)                       | Model-Monitoring-Lifecycle-SOP-v1.0.md §1, §4                                                  | <b>Implemented</b> |                                                               |
| <b>MG-1.3</b> | Responses to the AI risks deemed high priority, as identified by the MAP function, are developed, planned, and documented. Risk response options can include mitigating, transferring, avoiding, or accepting | Bias Program Layer 5 = mitigation via audit; PRD §16.10 + DPA §9.5.4 = mitigation via human-confirmation gates; no-demographic-ingest = avoidance; customer MSA no-sole-basis commitment = transfer-and-mitigate | Bias-Program-Framing-v1.0.md §§2, 6, 7; PRD V2 §16.10; EULA v2 DPA §9.5.4                      | <b>Implemented</b> |                                                               |
| <b>MG-1.4</b> | Negative residual risks (defined as the sum of all unmitigated risks) to both downstream acquirers of AI systems and end users are documented                                                                 | Bias Program §3.1 behavioral-proxy surface + §6.1.1 reconstruction risk + §6.4 defense-in-depth framing; Monitoring SOP "initial proposed — to validate against 90 days production data" threshold framing       | Bias-Program-Framing-v1.0.md §3.1, §6.1.1, §6.4; Model-Monitoring-Lifecycle-SOP-v1.0.md §1, §3 | <b>Implemented</b> | Unusually explicit acknowledgment for a venture-stage vendor. |
| <b>MG-2.1</b> | Resources required to manage AI risks are taken into account, along with viable non-AI alternative systems, approaches, or methods — to reduce the magnitude or                                               | Metadata-only mode is a non-content-analysis alternative; human-confirmation gates provide non-AI fallback; pricing tiers (Core / Complete) scope AI-integration depth                                           | PRD V2 §18.3; PRD V2 §16.10; Product/pricing-strategy-price-book.md                            | <b>Implemented</b> |                                                               |

| Action ID     | Action Family                                                                                                                                                                                                    | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                                                                                                      | Evidence Pointer                                                                                                | Maturity           | Notes                                                                                                                                                    |
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|               | likelihood of potential impacts                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                 |                    |                                                                                                                                                          |
| <b>MG-2.2</b> | Mechanisms are in place and applied to sustain the value of deployed AI systems                                                                                                                                  | Retraining triggers; gold-standard refresh; continuous monitoring                                                                                                                                                                                                                                                                                                                                            | Model-Monitoring-Lifecycle-SOP-v1.0.md §2, §4                                                                   | <b>Partial</b>     | Policy defines; first-cycle execution pending.                                                                                                           |
| <b>MG-2.3</b> | Procedures are followed to respond to and recover from a previously unknown risk when it is identified                                                                                                           | Monitoring SOP §6 incident runbook; model-lifecycle decision log; post-incident review; Bias Program §9.5 change management                                                                                                                                                                                                                                                                                  | Model-Monitoring-Lifecycle-SOP-v1.0.md §6; Bias-Program-Framing-v1.0.md §9.5                                    | <b>Implemented</b> |                                                                                                                                                          |
| <b>MG-2.4</b> | Mechanisms are in place and applied, and responsibilities are assigned and understood, to supersede, disengage, or deactivate AI systems that demonstrate performance or outcomes inconsistent with intended use | Model rollback runbook (K.5 Phase 4 deliverable); EULA v2 §7.2 advance-notice commitment for model updates; retraining-or-document-accept-with-risk decision pattern                                                                                                                                                                                                                                         | Model-Monitoring-Lifecycle-SOP-v1.0.md §4 (retraining triggers); Valley Bank Master Plan K.5                    | <b>Partial</b>     | Rollback-runbook pending (K.5 Phase 4 artifact); structural commitment exists via retraining-trigger decision pattern. Verified identifier per §5 below. |
| <b>MG-3.1</b> | AI risks and benefits from third-party resources are regularly monitored, and risk controls are applied and documented                                                                                           | Sub-processor due-diligence regime (EULA v2 §4, DPA §6); Vendor Risk Register v1 (K.10 Phase 2 deliverable)                                                                                                                                                                                                                                                                                                  | EULA v2 §4; DPA §6                                                                                              | <b>Partial</b>     | Contractual regime Implemented; published Vendor Risk Register Partial.                                                                                  |
| <b>MG-3.2</b> | Pre-trained models which are used for development are monitored as part of AI system regular monitoring and maintenance                                                                                          | Model-lifecycle decision log captures foundation-model version and change events; Monitoring SOP §4 output drift signals apply across pipeline; AI Supply Chain Posture Document (K.10 deliverable) scopes foundation-model deprecation runbook. The pre-trained model consumed for Tier-2 severity judgment varies by mode (Mode (a)/(b)/(c) per Architecture Lock Memo §D1); under Mode (a) the customer's | Model-Monitoring-Lifecycle-SOP-v1.0.md §4; Valley Bank Master Plan K.10; <b>Architecture Lock Memo v1.0 §D1</b> | <b>Partial</b>     | Verified identifier per §5 below. Foundation-model-specific monitoring responsibility tracks the mode; AXIA-owned output-drift monitoring is cross-mode. |

| Action ID     | Action Family                                                                                                                                                                                                                                         | AXIA Control / Artifact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Evidence Pointer                                                                                                                                | Maturity                                            | Notes                                                                                                                                                                                                                                                 |
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|               |                                                                                                                                                                                                                                                       | contracted managed-LLM service is responsible for model-version management; under Mode (b) the customer as endpoint operator is responsible; under Mode (c) AXIA is responsible for Llama 3.3 70B default (alternate open-weights swap path). AXIA's output-drift monitoring applies across all modes for the Gatekeeper and supervised-learning tiers.                                                                                                                                                                                                                                                  |                                                                                                                                                 |                                                     |                                                                                                                                                                                                                                                       |
| <b>MG-4.1</b> | Post-deployment AI system monitoring plans are implemented, including mechanisms for capturing and evaluating input from users and other relevant AI actors, appeal and override, decommissioning, incident response, recovery, and change management | Monitoring SOP §§1–8 is the post-deployment plan; feedback loop §8; incident runbook §6; change management (SDLC §6, Bias Framing §9.5). Post-deployment monitoring for the Tier-2 inference endpoint is mode-dependent: under Modes (a)/(b) endpoint-availability and inference-latency monitoring sits with the endpoint operator (customer or customer's contracted cloud provider); under Mode (c) AXIA operates the endpoint and monitors accordingly. AXIA's Gatekeeper and supervised-learning monitoring (gate FN rate, per-tier recall, output drift, analyst override) apply across all modes. | Model-Monitoring-Lifecycle-SOP-v1.0.md §§1–8; SDLC-Policy-v1.0.md §6; Bias-Program-Framing-v1.0.md §9.5; <b>Architecture Lock Memo v1.0 §D1</b> | <b>Implemented (policy) / Partial (operational)</b> | Verified identifier per §5 below. Policy comprehensive; operational execution pending 90-day calibration window and monitoring-dashboard delivery. Monitoring-responsibility allocation tracks the mode per Lock Memo §D1 AI risk-surface allocation. |
| <b>MG-4.2</b> | Measurable activities for continual improvements are integrated into AI system updates and include regular engagement with interested parties, including relevant AI actors                                                                           | Gold-standard refresh; post-incident-review monitoring improvements; semi-annual customer updates on Bias Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Model-Monitoring-Lifecycle-SOP-v1.0.md §6 Step 6; Bias-Program-Framing-v1.0.md §6.2                                                             | <b>Partial</b>                                      |                                                                                                                                                                                                                                                       |

| Action ID     | Action Family                                                                                                                                                                                            | AXIA Control / Artifact                                                                                                                                                    | Evidence Pointer                             | Maturity                                                     | Notes                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>MG-4.3</b> | Incidents and errors are communicated to relevant AI actors, including affected communities. Processes for tracking, responding to, and recovering from incidents and errors are followed and documented | DPA §7 incident notification (2 hours for regulated FIs); Schedule C §C.4 P0/P1/P2/P3 taxonomy; Schedule C §C.9 escalation path; Schedule C §C.10 monthly incident summary | EULA v2 DPA §7; Schedule C §C.4, §C.9, §C.10 | <b>Implemented (contractual) / Partial (operational SOP)</b> | Contractual commitments Implemented; Incident Response SOP v1 is K.5 Phase 4 deliverable (Partial). |

### 3. Generative AI Profile (NIST AI 600-1) — GenAI-Specific Actions

The NIST Generative AI Profile (July 2024) enumerates risks and risk categories specific to generative AI systems. AXIA's LLM-gating layer and LLM-based intel/advisor surface fall within GenAI Profile scope; the severity-classification ensemble (SetFit / ModernBERT / NN-XGBoost) is supervised-learning and maps primarily to NIST AI RMF 1.0 Core. The table below addresses Profile-specific risks that apply to AXIA's LLM-gating and intel surfaces.



| GenAI Risk / Action Family                                     | AXIA Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Evidence Pointer                                                                                                                               | Maturity                                                         | Notes                                                                                                                                                   |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Confabulation / hallucination</b>                           | Source Grounding principle: Intel responses reference actual data; no fabrication; factor-attribution explainability; structured WHO/WHAT/WHO rubric constrains LLM output                                                                                                                                                                                                                                                                                                                                                                                                                           | PRD V2 §16.8 (Source Grounding); <a href="#">severity-classification-framework.md</a>                                                          | <b>Implemented</b>                                               |                                                                                                                                                         |
| <b>Dangerous, violent, or hateful content</b>                  | Content Filtering Safety Guardrail: AI responses filtered for inappropriate content, bias, and discriminatory language                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PRD V2 §16.10 Safety Guardrails                                                                                                                | <b>Implemented</b>                                               |                                                                                                                                                         |
| <b>Data privacy (training data, prompt data, output data)</b>  | No training on customer data (DPA §9.5.1); cross-tenant isolation (DPA §9.5.2); data residency inside customer cloud tenant (DPA §4); data-handling terms for the Tier-2 inference endpoint vary by mode per Architecture Lock Memo §D1 — Mode (a) governed by the customer's service contract with the managed-LLM provider; Mode (b) entirely within the customer estate; Mode (c) AXIA-provided with documented data-handling posture per the Mode (c) one-pager. The Gatekeeper and full system run locally under customer control with no egress across all modes.                              | EULA v2 DPA §9.5.1, §9.5.2, §4; §1.4, §6.1 (cloud provider not sub-processor for Private/On-Prem); <b>Architecture Lock Memo v1.0 §D1, §D2</b> | <b>Implemented</b>                                               | Differentiator for financial-institution customers. Data-privacy surface for Tier-2 inference is resident with the endpoint operator per Lock Memo §D1. |
| <b>Environmental impact</b>                                    | Not assessed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | —                                                                                                                                              | <b>Planned (target Q4 2026)</b>                                  |                                                                                                                                                         |
| <b>Harmful bias and homogenization</b>                         | Bias Program Framing Layers 1–4 (including 3-vendor synthetic-data generation anti-homogenization control); Layer 5 (2026) population-level audit. Bias profile of the Tier-2 inference endpoint tracks the mode (Lock Memo §D1): Mode (a) inherited from customer's cloud-managed LLM; Mode (b) owned by customer as endpoint operator; Mode (c) inherited from published base model (Llama 3.3 70B default or alternate open-weights), no fine-tuning on customer data by default. AXIA bias controls apply to prompt construction, Gatekeeper prompt logic, and post-hoc review across all modes. | <a href="#">Bias-Program-Framing-v1.0.md</a> §§2–6; <b>Architecture Lock Memo v1.0 §D1</b>                                                     | <b>Partial (Layers 1–4 Implemented; Layer 5 Planned Q3 2026)</b> | Maps to MS-2.11.                                                                                                                                        |
| <b>Human-AI configuration (over-reliance, automation bias)</b> | Human-confirmation gates on all AI-suggested actions (PRD V2 §16.10); DPA §9.5.4 no-automated-decisions; Bias Framing §5 limits-of-human-                                                                                                                                                                                                                                                                                                                                                                                                                                                            | PRD V2 §16.10; EULA v2 DPA §9.5.4; <a href="#">Bias-Program-Framing-v1.0.md</a> §5                                                             | <b>Implemented (structural) / Partial</b>                        |                                                                                                                                                         |

| GenAI Risk / Action Family                                       | AXIA Control                                                                                                                                                                                                                                                                                                                                               | Evidence Pointer                                                                  | Maturity                  | Notes                                                                                       |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------|
|                                                                  | confirmation acknowledgment + Layer 5 human-override-rate analysis scope                                                                                                                                                                                                                                                                                   |                                                                                   | (override-rate telemetry) |                                                                                             |
| Information integrity                                            | Audit log (PRD V2 §21.4); model-version capture per decision; change logs in SDLC Policy + Monitoring SOP                                                                                                                                                                                                                                                  | PRD V2 §21.4; SDLC-Policy-v1.0.md §6, §9; Model-Monitoring-Lifecycle-SOP-v1.0.md  | Partial                   | 100% audit-log-coverage target not yet attested (see MS-2.8).                               |
| Information security                                             | EULA v2 §7.4 security incident notification; DPA §4 data residency; cross-tenant isolation; red-team quarterly cadence includes prompt-injection testing                                                                                                                                                                                                   | EULA v2 §7.4; DPA §4; Model-Monitoring-Lifecycle-SOP-v1.0.md §7                   | Partial                   | First red-team exercise pending.                                                            |
| Intellectual property                                            | No training on customer data (DPA §9.5.1); sub-processor flow-down (DPA §6.4.1); foundation-model IP posture varies by mode per Architecture Lock Memo §D1 — Modes (a)/(b) governed by customer's service contract with the managed-LLM provider; Mode (c) Llama 3.3 70B default license terms apply                                                       | EULA v2 DPA §9.5.1, §6.4.1; Architecture Lock Memo v1.0 §D1                       | Implemented               | IP-surface allocation tracks the mode.                                                      |
| Obscene, degrading, and/or abusive content                       | Content Filtering Safety Guardrail                                                                                                                                                                                                                                                                                                                         | PRD V2 §16.10                                                                     | Implemented               |                                                                                             |
| Toxicity                                                         | Content Filtering Safety Guardrail; WHO/WHAT/WHO adjudication rubric consistent                                                                                                                                                                                                                                                                            | PRD V2 §16.10; severity-classification-framework.md                               | Implemented               |                                                                                             |
| Value chain and component integration (third-party models, data) | EULA v2 §4 sub-processor regime; AI Supply Chain Posture Document (K.10 Phase 2 deliverable); foundation-model inventory varies by mode per Architecture Lock Memo §D1 — Modes (a)/(b) use customer-selected provider under customer service contract; Mode (c) uses AXIA-selected Llama 3.3 70B default with documented alternate-open-weights swap path. | EULA v2 §4; DPA §6; Valley Bank Master Plan K.10; Architecture Lock Memo v1.0 §D1 | Partial                   | AI Supply Chain Posture Document pending. Value-chain component allocation tracks the mode. |

#### 4. Summary Table — Maturity Snapshot

Counts are across all rows in §§2–3 (total 70 mapped entries). Rows marked with combined tags (e.g., "Implemented (policy) / Partial (execution)") are counted under the weaker of the two tags for honesty.

| NIST Function / Profile                         | Implemented | Partial   | Planned   | Total Rows |
|-------------------------------------------------|-------------|-----------|-----------|------------|
| <b>Govern</b>                                   | 6           | 4         | 6         | 16         |
| <b>Map</b>                                      | 10          | 3         | 2         | 15         |
| <b>Measure</b>                                  | 5           | 12        | 1         | 18         |
| <b>Manage</b>                                   | 3           | 6         | 0         | 9          |
| <b>Govern / Map / Measure / Manage subtotal</b> | <b>24</b>   | <b>25</b> | <b>9</b>  | <b>58</b>  |
| <b>Generative AI Profile</b>                    | 7           | 4         | 1         | 12         |
| <b>TOTAL</b>                                    | <b>31</b>   | <b>29</b> | <b>10</b> | <b>70</b>  |

**Honest read:** AXIA is strongest on Map (context, purpose, privacy risk framing — 10 Implemented) and on structural contract / privacy controls that cut across functions. AXIA is weakest on Govern (formal AI Policy, risk-tolerance statement, diverse team, personnel training — all Planned) and on Measure operational execution (policy is written in Monitoring SOP v1.0, but continuous-cadence metrics and first-run instances depend on dashboard delivery and the 90-day calibration window). This is the correct venture-stage pattern — structural and contractual controls arrive first; operational artifacts and independent attestation follow with scale.

## 5. NIST Action-ID Verification on Bias Program Framing Citations

Bias Program Framing v1.0 §8 cited preliminary action-IDs as "mapping hypotheses pending verification during Phase 2.1 NIST Crosswalk." This Crosswalk is that verification. The table below records the verification outcome for each action-ID the Bias Framing cited.

| Action-ID cited in Bias Framing                                    | Bias Framing usage                                                               | Verification outcome                                                                                                                                                                                                                                                                                                                                                                                                                      | Notes                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GV-1.1</b> (bias policy / legal-regulatory AI requirements)     | Layer 1 — Input Minimization policy                                              | <b>Verified at family level; requires NIST published-text cross-reference for exact sub-action identifier.</b> The family "Legal and regulatory requirements regarding AI are understood, managed, and documented" is consistent with the NIST AI RMF 1.0 Core Govern category structure. The exact wording "GV 1.1" vs. "Govern 1.1" and any sub-action decimal (e.g., GV-1.1.1) requires cross-reference.                               | The Bias Framing usage is consistent with the family's intent. Flag: verify exact identifier text against NIST AI RMF 1.0 Core.                                                                                                                                             |
| <b>MS-2.11</b> (harmful bias measurement)                          | Layer 2 — Engineering Diversity + Layer 5 — Population-Level Audit               | <b>Verified at family level; the "harmful bias and homogenization" action family is canonical in NIST AI RMF 1.0 Core Measure function.</b> Mapping to AXIA's 3-model ensemble + 3-vendor synthetic data as anti-homogenization control is appropriate.                                                                                                                                                                                   | Strongest of the Bias Framing identifiers; family-level content matches. Verify exact identifier numbering vs. NIST published text.                                                                                                                                         |
| <b>MG-3.2</b> (bias monitoring / pre-trained model monitoring)     | Layer 4 — Explainability substrate; Layer 5 — roadmap                            | <b>Verified at family level.</b> The family "Pre-trained models which are used for development are monitored as part of AI system regular monitoring and maintenance" is canonical. Bias Framing use applied it to bias-monitoring substrate via explainability outputs — this is a <b>reasonable but less-than-exact fit</b> . MG-3.2 is foundation-model-monitoring specific; bias monitoring more precisely maps to MS-2.11 + MG-4.1.  | <b>Finding:</b> Bias Framing §8 row "Layer 4 — MG-3.2 (bias monitoring)" should be revised in next Bias Framing version to reference <b>MS-2.11 + MG-4.1</b> for bias monitoring; MG-3.2 retained specifically for foundation-model monitoring. Note for Bias Framing v1.1. |
| <b>GV-4.1</b> (accountable team for bias)                          | Layer 3 — Customer-Configurable Guardrails ownership                             | <b>Verified at family level.</b> "Organizational policies and practices are in place to foster a critical thinking and safety-first mindset" is canonical. Mapping to named Content Filtering / bias governance / escalation ownership is appropriate.                                                                                                                                                                                    |                                                                                                                                                                                                                                                                             |
| <b>MP-2.3 / MP-4.1</b> (context and risk framing)                  | Layer 1 — purpose limitation framing                                             | <b>Verified at family level.</b> MP-2.3 (scientific integrity and TEVV considerations) and MP-4.1 (third-party data/software legal risks) are both canonical. MP-2.3 fits AXIA's TEVV substrate (severity benchmark + framework); MP-4.1 fits sub-processor + foundation-model context. <b>MP-2.3 is the better fit for the Bias Framing usage; MP-4.1 is better recast as sub-processor / third-party risk rather than bias framing.</b> | <b>Finding:</b> Bias Framing §8 row "Layer 1 — MP-2.3 / MP-4.1 (context and risk framing)" — retain MP-2.3; consider moving MP-4.1 to a sub-processor or GenAI-Profile foundation-model row in the next Bias Framing version. Note for Bias Framing v1.1.                   |
| <b>MG-4.1</b> (model risk management / post-deployment monitoring) | Layer 2 — Engineering Diversity confidence-weighted voting + adjudication rubric | <b>Verified at family level.</b> MG-4.1 "Post-deployment AI system monitoring plans are implemented" is canonical. Bias Framing used it specifically for ensemble voting + rubric; this is a reasonable fit but MG-4.1 is a broader operational-monitoring control. <b>Better primary mapping for Layer 2 engineering-diversity: MS-2.11 (anti-homogenization) as the canonical measurement control; MG-4.1 as secondary.</b>             | <b>Finding:</b> Bias Framing §8 row "Layer 2 — MG-4.1 (model risk management)" — retain MG-4.1 as secondary; elevate MS-2.11 as primary. Note for Bias Framing v1.1.                                                                                                        |

| Action-ID cited in Bias Framing                                            | Bias Framing usage                          | Verification outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Notes                                                                                                                                                                 |
|----------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MS-2.8 / MS-2.9</b><br>(explainability / interpretability)              | Layer 4 — Explainability                    | <b>Verified at family level.</b> Both are canonical NIST AI RMF 1.0 Core Measure actions. MS-2.8 (transparency and accountability risks) and MS-2.9 (AI model is explained, validated, documented) both map to PRD V2 §16.8 + §21.4. <b>Mapping is correct.</b>                                                                                                                                                                                                                          |                                                                                                                                                                       |
| <b>GV-1.3 / GV-5.1</b><br>(bias policy enforcement / stakeholder feedback) | Layer 5 — audit cadence + customer delivery | <b>Partially verified.</b> GV-5.1 (feedback from those external to the team) is canonical and fits customer-delivery cadence. GV-1.3 (risk-tolerance-based process determination) is canonical but is a <b>weaker fit</b> for "bias policy enforcement"; GV-1.3 is about risk-tolerance-driven process scaling, not policy enforcement specifically. <b>Better mapping for bias policy enforcement is GV-1.1 (already cited in Layer 1) + GV-5.2 (adjudicated feedback integration).</b> | <b>Finding:</b> Bias Framing §8 row "Layer 5 — GV-1.3 / GV-5.1" — retain GV-5.1; replace GV-1.3 with GV-5.2 in next Bias Framing version. Note for Bias Framing v1.1. |

### 5.1 Aggregate Verification Status

- **Identifiers verified at family level (consistent with canonical NIST AI RMF 1.0 structure):** All nine cited identifiers (GV-1.1, MS-2.11, MG-3.2, GV-4.1, MP-2.3, MP-4.1, MG-4.1, MS-2.8, MS-2.9, GV-1.3, GV-5.1).
- **Identifiers requiring Bias Framing v1.1 refinement for more precise primary mapping:** MG-3.2 (shift to MS-2.11 + MG-4.1 for bias monitoring), MP-4.1 (shift to sub-processor context), MG-4.1 (secondary rather than primary for Layer 2), GV-1.3 (replace with GV-5.2 for policy enforcement).
- **Identifiers requiring NIST published-text cross-reference for exact sub-action decimal identifier text:** All nine. This is a Cannot Assess Without item in §7 below.

No identifier cited in Bias Framing v1.0 §8 has been verified as incorrect or fabricated. The refinements above are precision adjustments for Bias Framing v1.1, not corrections.

## 6. Findings

### Finding 1

- **What:** Summary table shows 31 Implemented / 29 Partial / 10 Planned across 70 rows. Partial is the largest bucket — driven by Monitoring SOP policy being approved but operational execution (continuous-cadence metrics, first red team, analyst-override instrumentation, gold-standard first refresh) being gated on monitoring dashboard delivery and the 90-day calibration window.
- **Why it matters:** Valley's MRM reviewers will read "Partial" as an honest venture-stage posture, but the volume of Partial rows concentrated in Measure is a legitimate maturity signal. The mitigation is visible commitment to the Phase 1 P0 items (dashboard, analyst-override instrumentation) with dated milestones in the Valley Bank submission cover note.
- **Severity:** P1
- **Suggested next step:** Surface Phase 1 P0 items in the Valley submission cover note with committed delivery windows from the P1 effort plan. Do not upgrade any Partial row to Implemented until the operational artifact is live.

### Finding 2

- **What:** Govern function is 6 Implemented / 4 Partial / 6 Planned. The Planned items (formal written AI Policy GV-1.2; risk-tolerance statement GV-1.3 / MP-1.5; diverse team GV-3.1 / MP-1.2; personnel training GV-2.2; environmental impact MS-2.12) are correctly mapped to venture-stage absence.

- **Why it matters:** Valley MRM will notice the Govern concentration. Honest framing (venture-stage, post-SOC-2 scoping) is defensible; overclaim would be disqualifying.
- **Severity:** P1
- **Suggested next step:** Cover note to Valley explicitly acknowledges Govern-function maturity as venture-stage and commits to SOC 2 Type II Q4 2026 as the milestone at which formal policy and risk-tolerance artifacts are produced. Asaf + @general-counsel sign off on the commitment framing.

### Finding 3

- **What:** MS-2.8 (100% audit-log-coverage target) is tagged Partial because PRD V2 §21.4 carries it as a Trust Metric **target**, not an attested production reality. Cannot Assess Without Engineering Confirmation (carried from Bias Program Framing Finding).
- **Why it matters:** If Dan and Orr confirm 100% coverage is live in production today, MS-2.8 upgrades to Implemented and MS-2.4 benefits. If it remains aspirational, the Crosswalk should say so explicitly in both rows.
- **Severity:** P1
- **Suggested next step:** Dan and Orr provide written attestation of current audit-log coverage percentage. If < 100%, Crosswalk updated to show current percentage with target-date commitment.

### Finding 4

- **What:** Bias Framing v1.0 §8 action-ID mapping has four refinements identified in §5 above (MG-3.2 → MS-2.11 + MG-4.1 for bias monitoring; MP-4.1 context-framing recast; MG-4.1 as secondary for Layer 2; GV-1.3 → GV-5.2 for policy enforcement).
- **Why it matters:** Bias Framing is a signed deliverable. The refinements are precision adjustments, not corrections — no identifier was wrong or fabricated. Propagating the refinements into Bias Framing v1.1 is a maintenance task, not a gate.
- **Severity:** P2
- **Suggested next step:** Open Bias Framing v1.1 work item on next scheduled revision cycle. No immediate action needed for Valley submission; the current Bias Framing §8 row prose is defensible at family level.

### Finding 5

- **What:** All cited action-ID decimal identifiers (GV-1.1, MS-2.11, etc.) are verified at family level against the canonical NIST AI RMF 1.0 Core structure; exact sub-action decimal identifiers (e.g., whether "GV 1.1" is rendered as "GV-1.1" or "GV.1.1" or "Govern 1.1") require cross-reference against the NIST published text.
- **Why it matters:** A bank MRM reader who looks up a specific identifier in the NIST publication wants exact textual alignment, not family-level paraphrase.
- **Severity:** P1 — Pass-2 Blocker
- **Suggested next step:** @compliance-officer or @general-counsel performs a one-pass cross-reference against NIST AI RMF 1.0 Core publication (NIST AI 100-1, January 2023) + NIST Generative AI Profile (NIST AI 600-1, July 2024) text before Valley submission. Edit exact identifier text where format differs; do not change the family-level mapping logic.

### Finding 6

- **What:** GenAI Profile coverage (§3) is 7 Implemented / 4 Partial / 1 Planned across 12 Profile-specific risks. Environmental impact is the sole Planned row.
- **Why it matters:** Valley's MRM may ask whether AXIA has assessed environmental impact of model training and inference. Venture-stage absence is defensible but worth flagging in the cover note to avoid a diligence surprise.
- **Severity:** P2
- **Suggested next step:** Brief cover-note acknowledgment of environmental-impact assessment as Q4 2026 planned artifact.

### Finding 7

- **What:** Several rows use combined tags ("Implemented (policy) / Partial (execution)"). The summary table in §4 counts these under the weaker tag for honesty.

- **Why it matters:** A reader scanning only §4 does not see the "policy vs execution" distinction. The distinction is material — policy-complete is a credible venture-stage posture even before operationalization.
- **Severity:** P2
- **Suggested next step:** Valley submission cover note frames Partial rows in two categories: "policy Implemented, execution pending operationalization" vs "structural commitment, full artifact pending." This reframe helps a bank reviewer understand the venture-stage posture.

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## 7. Reviewer Checklist

Reviewers sign off in order.

- [ ] **Yohay Etsion (compliance sponsor):** Framing Statement (§1) posture accepted. Three-state maturity tagging applied consistently (no row upgraded to Implemented without an evidence pointer). Govern-function honest positioning (§1.3) accepted. Findings 1–7 triaged.
- [ ] **Dan (AI lifecycle owner):** §2.2 MAP rows on TEVV (MP-2.3), architecture (MP-2.1), safety / guardrails (MP-1.6, MP-2.2) accurately reflect production AI architecture. §2.3 MEASURE rows on monitoring metrics (MS-1.1, MS-2.3, MS-2.4) and fairness / bias (MS-2.11) accurately reflect Monitoring SOP v1.0 and Bias Program Framing v1.0 postures. §3 GenAI Profile rows on confabulation, harmful bias, human-AI configuration accurately reflect current guardrail implementation. Finding 3 attestation on current audit-log coverage percentage provided (or target commitment logged).
- [ ] **Orr (Head of Engineering):** §2.3 MS-1.1 + MS-2.4 monitoring-dashboard dependency correctly reflected as Partial. §2.4 MG-4.1 + MG-4.3 operational-execution dependency correctly reflected. Finding 1 P0 dependency (dashboard delivery + analyst-override instrumentation) commitment and timeline confirmed.
- [ ] **@ai-architect (Pass 2 controls mapping accuracy):** Controls mapped to NIST Functions / Actions are technically coherent with AXIA's V4 pipeline, ensemble architecture, and LLM gating surfaces. GenAI Profile (§3) coverage is appropriate for AXIA's LLM-gating and Intel surfaces; supervised-learning ensemble correctly routed to Core rather than Profile. Action-ID refinements in §5 verified as correct precision adjustments.
- [ ] **@chief-architect (Pass 2 independence verification):** Crosswalk does not overclaim. No row tagged Implemented without an evidence pointer. Govern-function Planned rows not shaded into Implemented. Partial-heavy Measure function honestly reflects operational-execution gap. Summary table (§4) counts are consistent with row-level tags (weaker-of-two rule applied).
- [ ] **@general-counsel (Pass 2 regulatory framing):** Disclaimer block verbatim from sensitive-skill-guardrails.md §3.1; jurisdiction declared. §1 explicit "AXIA does not claim NIST AI RMF conformance" framing accepted. §1 "designed to support customer SR 11-7 obligations" framing accepted — no "AXIA is SR 11-7 compliant" language anywhere in the document. §5.1 action-ID verification status does not overclaim NIST alignment. §6 Finding 5 cross-reference commitment against NIST published text before Valley submission understood and scheduled.
- [ ] **All P0 and P1 findings** (Findings 1, 2, 3, 5 = P1; no P0 surfaced in Crosswalk itself; Finding 1 carries Phase 1 P0 dashboard dependency) addressed or explicitly accepted as risk.
- [ ] **Action-ID cross-reference complete (Finding 5)** against NIST AI RMF 1.0 Core (NIST AI 100-1, January 2023) + NIST Generative AI Profile (NIST AI 600-1, July 2024). Identifier format verified; no identifier remains in the document that cannot be verified against NIST published text.
- [ ] **Valley submission cover note** drafted with Finding 1 P0-dependency framing, Finding 2 Govern-maturity framing, Finding 6 environmental-impact acknowledgment, and Finding 7 Partial-row reframe.

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## 8. Cannot Assess Without

Items deliberately not opined on in this Crosswalk. Each requires external or specialist verification before the document is relied on in a binding submission.

- **Cannot Assess Without — NIST Publication Verification:**



- **Exact NIST action-ID decimal-identifier text** for every identifier cited in §§2–3 (GV-1.1 through MG-4.3 and Profile rows). Verified at family level; exact rendering ("GV-1.1" vs. "Govern 1.1" vs. "GV.1.1") and exact sub-action decimal identifiers (e.g., whether GV-1.1 is a leaf action or has sub-actions GV-1.1.1, GV-1.1.2) require one-pass cross-reference against NIST AI RMF 1.0 Core (NIST AI 100-1) and NIST Generative AI Profile (NIST AI 600-1) published text. Finding 5 scopes the work; not complete in v1.0.
- **Generative AI Profile action-ID mapping** for §3 rows. NIST AI 600-1 presents Profile content as risk categories with suggested actions cross-referenced to Core actions; the mapping in §3 is at the risk-category level and does not cite Profile-specific action identifiers. If Valley requests Profile-specific identifiers, a second pass against NIST AI 600-1 is required.
- **Cannot Assess Without — Engineering Confirmation:**
  - **Current audit-log coverage percentage in production** (PRD V2 §21.4 target is 100%; current attained percentage is not attested in this Crosswalk). Dan + Orr attestation required. Affects MS-2.8 maturity tag.
  - **Whether monitoring-dashboard delivery and analyst-override instrumentation** have landed (P1 effort plan). Affects MS-1.1, MS-2.4, GV-5.1, MG-4.1 maturity tags. If delivery shifts, Partial rows remain Partial; if delivery lands, specific rows upgrade to Implemented.
  - **Whether V5 / V6 of the severity pipeline has superseded V4 numbers** (QWK 0.920, ModernBERT 80.4%, per-tier thresholds). If V5/V6 has materially changed numbers, MS-2.3 and MS-2.5 rows may need restatement; threshold baselines in Monitoring SOP (and referenced here) may be understated.
- **Cannot Assess Without — Licensed Counsel and Specialist Review:**
  - **Regulatory framing consistency across Valley submission.** This Crosswalk's "designed with reference to NIST AI RMF" and "designed to support customer SR 11-7 obligations" framing must be consistent with the K.5 Incident Response SOP response, the K.8/K.9 Bias Framing response, the K.7 DPA Schedule A-US draft, and the K.10 Vendor Risk Register response. @general-counsel owns consistency review across the set.
  - **Whether Valley's MRM function expects mapping to any additional frameworks** beyond NIST AI RMF 1.0 + GenAI Profile (e.g., OCC 2013-29 specifically, FFIEC IT Handbook, NYDFS Part 500 §500.11, ISO/IEC 42001, EU AI Act Annex III if Valley has EU operations). If yes, additional Crosswalk appendices required.
  - **Whether any Implemented-tagged row would be challenged as overclaim under Valley's MRM effective-challenge process.** Every "AXIA does X" claim in the matrix must survive Valley diligence. This Crosswalk is a drafting aid; submitted Crosswalk is a commitment.
- **Cannot Assess Without — Customer MRM Integration:**
  - **Customer MRM Integration Guide format** (how Valley's MRM team requests and receives the artifacts listed in Bias Program Framing §7.2 and this Crosswalk's Implemented rows). Deferred to post-Valley-response-submission.
  - **Whether Valley's vendor-risk team maintains a crosswalk template** AXIA should align to rather than supplying its own format. If yes, reformat required.

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## 9. Document Control

- Path: `AXIA/Product/compliance/NIST-AI-RMF-Crosswalk-v1.0.md`
- Supersedes: None (first version)
- Next review: Event-triggered — on any NIST AI RMF Core or Profile version update; on any material AXIA model or architecture change (per SDLC Policy §6 Change Classes); on Valley MRM challenge producing requested revisions. Annual review otherwise (2027-04-20).
- Canonical master plan: `AXIA/Product/valley-bank-questionnaire-master-response-plan.md`
- Phase 2 deliverable per master plan
- Source evidence:
- PRD V2: §16.8, §16.9, §16.10, §18.3, §21.4
- AI architecture: `axia-ai-system-analysis.md`, `v4-pipeline-analysis.md`, `severity-classification-framework.md`, `severity-benchmark-analysis.md`, `severity-benchmark-detailed-gt.md`



- Phase 1 policies: `AXIA/Product/policies/SDLC-Policy-v1.0.md` , `AXIA/Product/policies/Model-Monitoring-Lifecycle-SOP-v1.0.md` , `AXIA/Product/policies/Bias-Program-Framing-v1.0.md`
- Contractual: EULA v2 §4, §7.2, §7.4; DPA §4, §6, §7, §9.5.1, §9.5.2, §9.5.3, §9.5.4, §11; Schedule C §C.4, §C.9, §C.10
- V4 numbers canonical: `AXIA/Product/valley-bank-v4-numbers-canonical.md`
- NIST publications (for cross-reference per Finding 5): NIST AI RMF 1.0 Core (NIST AI 100-1, January 2023); NIST Generative AI Profile (NIST AI 600-1, July 2024)